

Dry.pdf Matam HW 3

1. Requirements for T type in order to use our generic SortedList template
 - a. The most critical requirement for T is that it has a > method for comparing with other T types – without this our insert function will not work
 - b. Constructor – every T type needs to have a constructor
 - c. Copy Constructable – this is required for various constructors of SortedList
 - d. Assignable
 - e. Predicate compatible – type T must have Boolean functions that work for it, in order to use the filter function
 - f. Function compatibility – in order to use apply function
 - g. Destructor – every T type must have a destructor, as SortedList does not deal with deletion of any T type
 - h. Note: an earlier implementation of our SortedList code created default T values, using new Node<T> in numerous places. This was proven to be problematic because for when T = Task, Task.h has no default constructor, and so program ran into a compilation error whenever it tried to create a default Node<Task>. The instructions clearly forbid us from changing Task.h, but otherwise we would have added a default constructor (one without any variables)
2. Why we could not allow for a non-const iterator:
 - a. If we allowed for a non-const iterator, this would allow us to receive iterators and change their class data using methods available to the public. This would in turn allow us to access items in a SortedList, and change their inner values, which could change the order which they need to be placed in the list!
 - b. For example, with the TaskManager application, if we were to add to an “employee” a task that had the highest priority, we could then use an iterator to access the priority level of that task and make it the lowest priority. The SortedList would not change its location, and the order of the list would be ruined.
3. How to use filter to accomplish the functionality required by the student:
 - a. I would recommend using the same idea that we used in our project – create a static variable which I could then change during runtime, and bool functions that use that static variable